

Chapter 3

Business Cycle Measurement

■ Teaching Goals

Chapter 3 stresses the importance of observation as a foundation for scientific exploration in macroeconomics. Because there are mountains of data measurements about the macroeconomy, we need to begin to organize these data in a way that we can begin to look for regularities in the economy—regularities that we hope to explain. The cornerstone of business cycle analysis is deviations from trend in real GDP. Students must first understand the difference between long-run trends and deviations from trend in real GDP.

One of the most exciting things about the study of business cycles is the observation that each of these recurring cycles is not unique. Because successive business cycles are more alike than they are different, we have the hope of providing an explanation of business cycles that can consistently be applied to each new cycle we encounter. These business cycle regularities, the “stylized facts” of business cycles, provide us with the first clues about the nature of the typical cycle. Although students may take some time to digest and remember these facts, it is important to be clear about the motivation for cataloging these facts.

The plan of study of this text is to view the business cycle as a puzzle that we hope to solve. Our task is much the same as a detective trying to solve a murder mystery. The first task is to collect clues. The clues in the study of business cycles are the regularities chronicled in this chapter. Later on we will begin to encounter “suspect” causes of business cycles. In order to solve the mystery, we need to deduce the “who,” “what” and “why” of our mystery. We cannot begin the deduction process before first collecting the clues. Students need to understand this detective process. Otherwise, the clues presented in this chapter will be nothing more than a laundry list that they must memorize.

■ Classroom Discussion Topics

To get the ball rolling it might be useful to ask the students about whether they have concerns about recessions. Is the economy currently in a recession? When was the economy most recently in recession? How do we date recessions – turning points in macroeconomic activity? Should we be paying attention to only real GDP or to other indicators that might be important for economic welfare – e.g. employment or the unemployment rate?

Macroeconomics is primarily concerned with business cycles and economic growth. Ask the students which of these two topics is most important. A typical comparison is between the possibility of completely eliminating recessions as opposed to increasing the rate of growth by a half of one percent or so. If they could pick only one of these advances, which would they pick? Some simple calculations reveal the apparent dominance of more growth over eliminating business cycles. Ask them if they believe

that there is any other cost of a recession other than the value of lost GDP. In particular, emphasize that GDP fluctuations hide a lot about fluctuations at the household level.

Another potential discussion centers on the murder mystery analogy. If the students were rounding up the usual suspects of business cycle causes, which suspects would they pick? One part of detective work involves theories and hunches about what might have happened. Another less glamorous part of detective work involves ruling out some of the suspects. How can studying the clues eliminate a suspect from consideration? Emphasize the point that much of the organization of the text revolves about the possible elimination of suspected causes of business cycles based on the evidence.

■ **Outline**

- I. Regularities in GDP fluctuations**
- II. Co-movement**
 - A. Cyclical
 - B. Lead/Lag Relationships
 - C. Variability
- III. Components of GDP**
 - A. Consumption
 - 1. Procyclical
 - 2. Coincident
 - 3. Less Variable Than GDP
 - B. Investment
 - 1. Procyclical
 - 2. Residential and Inventory Investment Lead GDP
 - 3. More Variable Than GDP
- IV. Nominal Variables**
 - A. The Phillips Curve
 - B. Aggregate Price Level
 - 1. Procyclical Since World War II
 - 2. Sometimes Countercyclical
 - 3. Lags GDP
 - C. Money Supply
 - 1. Procyclical
 - 2. Leads GDP
- V. Labor Market Variables**
 - A. Employment
 - 1. Procyclical
 - 2. Lags GDP
 - B. Real Wage Rates
 - 1. Procyclical
 - 2. Aggregation Problems
 - C. Average Labor Productivity
 - 1. Procyclical
 - 2. Coincident
 - 3. Less Variable Than GDP
- VI. Seasonal Adjustment**

A. Seasonal Data

■ **Solutions to End-of-Chapter Problems**

1. In Figure 3.13, after 1980 it appears that real GDP and the nominal money supply are negatively correlated, and it is difficult to discern any lead/lag relationship between real GDP and nominal money. Many people, particularly old-school Monetarists (Milton Friedman included) ascribed much importance to the money-output correlation. The fact that this relationship appears to have disappeared in the last 30 years of data is also important. The reason we observe this could have something to do with changes in the way monetary policy was conducted after 1980.
2. In Figure 3.16, what we tend to see during the last 3 recessions is a quick recovery of average labor productivity prior to real GDP recovering to the point where it is above trend. This is part of the jobless recovery phenomenon. Employment has been very slow to recover during the last three recessions, which makes for higher growth in average labor productivity coming out of the recession. We get this as average labor productivity is the ratio of real GDP to employment, so if employment is not recovering and real GDP is, then the ratio shows a spurt of growth.
3. Both consumption and investment are procyclical and coincident. The key difference is that investment is much more volatile than consumption. Consumer durables provide services over a horizon greater than one year. Some consumer nondurable products, like apparel, provide services well beyond the date of purchase. Services, by definition, are fully utilized at the point of sale. The same kinds of timing considerations that affect business investment are likely to come into play with consumer durables, and to a lesser extent, consumer nondurables. Therefore, it is logical that consumer durable purchases should be more volatile than consumer nondurable purchases and that consumer nondurable purchases should be more volatile than consumption of services.
4. After the 1981-82, the cyclical behavior of the price level varies. From 1980-2000, the price level appears to be countercyclical, but after 2000 it appears to be procyclical. This is important, as alternative types of macroeconomic shocks will move the price level in different directions. As a result, the cyclicity of the price level may be useful in uncovering what shocks are driving aggregate fluctuations at particular dates.
5. One can see that the amplitude of the deviations from the trend is more moderate for prices, as for GDP. It appears also that the price level is becoming less countercyclical. The fact that both GDP and the price level changed their cyclical behavior indicates that any explanation of the Great Moderation needs to look at more than GDP.